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Assessment 3: Unsupervised Learning

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1 Introduction

Airbnb is a popular platform for short-term rentals for travelling tourists. Cape Town is a major tourist destination in South Africa, and many Cape Town homes are being rented out on Airbnb. Unsupervised learning, such as cluster analysis, is a useful technique to gain insight into the structure of a dataset. Airbnb provides information about properties being rented out on their platform. Broadly, the information available about Airbnb properties include the location, physical capacity, rental information and reviews.

An interesting area to explore about Airbnb properties in Cape Town is examining the guest experience across geographical zones. This report analyses the use of unsupervised learning techniques in better understanding Airbnb properties in Cape Town.

2 Exploratory Data Analysis

Airbnb dataset contains 40 variables about 16,317 Airbnb properties. The variables of interest about properties are about their physical capacity, guest reviews, location and price.

Exploratory Data Analysis on the dataset reveals that some properties have missing data relating to their physical capacity and reviews. The missing data was imputed with ward level medians and the neighbourhood names variable was dropped due to missing data and large amounts of generic labels.

Table 1: Tables of missing observations

Variable	Missing Observations
Neighbourhood	5344
Beds	14
Bedrooms	6
Review_Scores_Checkin	5
Review_Scores_Communication	5
Review_Scores_Location	5
Review_Scores_Value	5
Bathrooms	4
Review_Scores_Accuracy	4
Review_Scores_Cleanliness	4

By plotting histograms (Figure 1), we found that the dataset has a number of outliers on multiple variables of interest. To deal with the outliers, we log transformed the price variable and visually selected a threshold to cap and grouped extreme values together.

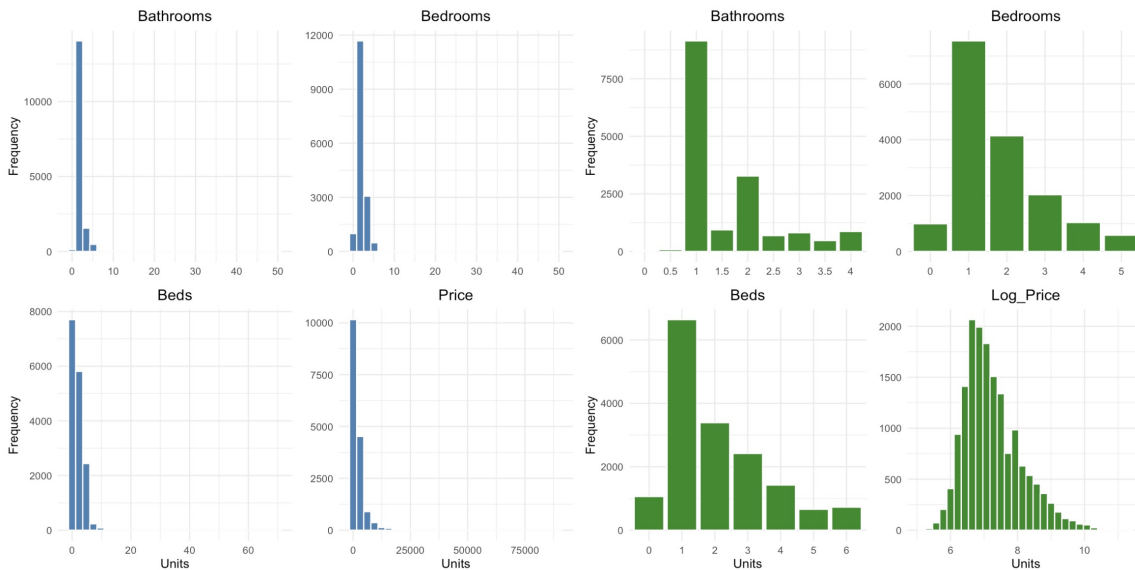


Figure 1: Histogram showing the distributions of physical attributes and price. Left plot has distributions which are right skewed as a result of outliers. Right plot shows distribution after transformation.

The correlation matrix was obtained for variables relating to physical capacity, guest reviews and price. The correlation matrix shows that physical capacity variables are highly correlated between themselves and review variables are highly correlated between themselves.

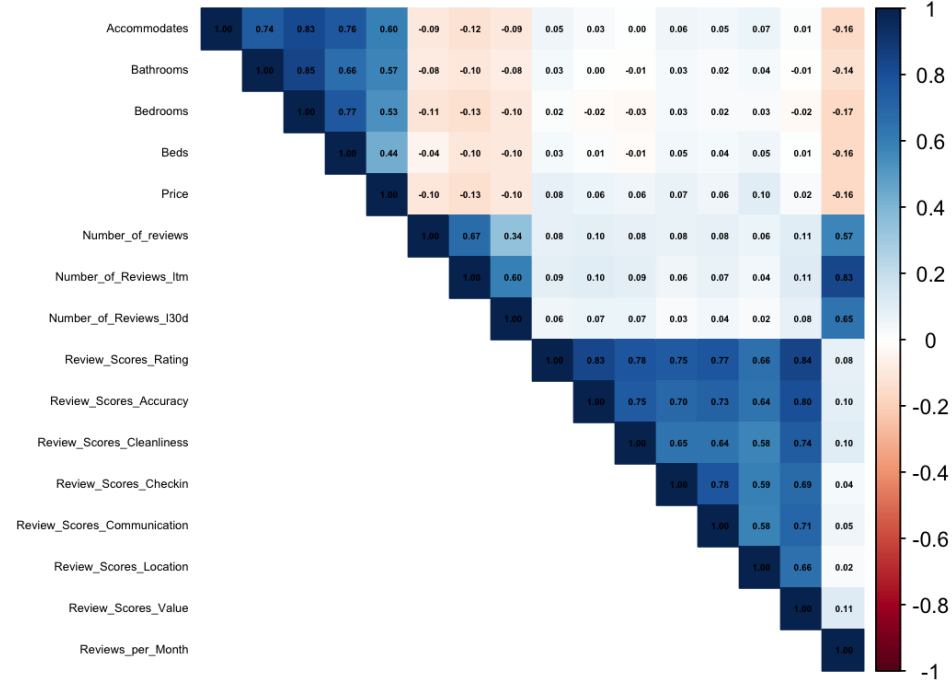


Figure 2: Correlation plot of variables

PCA was done separately on physical capacity and review variables (Figure 3). The first skree plot shows that PCA can combine variables into one variable relating to physical capacity. The second skree plots shows that review variables can be combined into two PCA's relating to review quality and the quantity of reviews. Hence, the dimension of the data can be reduced to 1 PC component representing physical capacity and 2 PC components representing review quality and volume. Ward level average also calculated to embedded spatial information and average price. Room_Type was used in favour of Property_Type due the variable being a fine grained representation of Room_Type. Variables that are used further in the analysis are shown in Table 2.

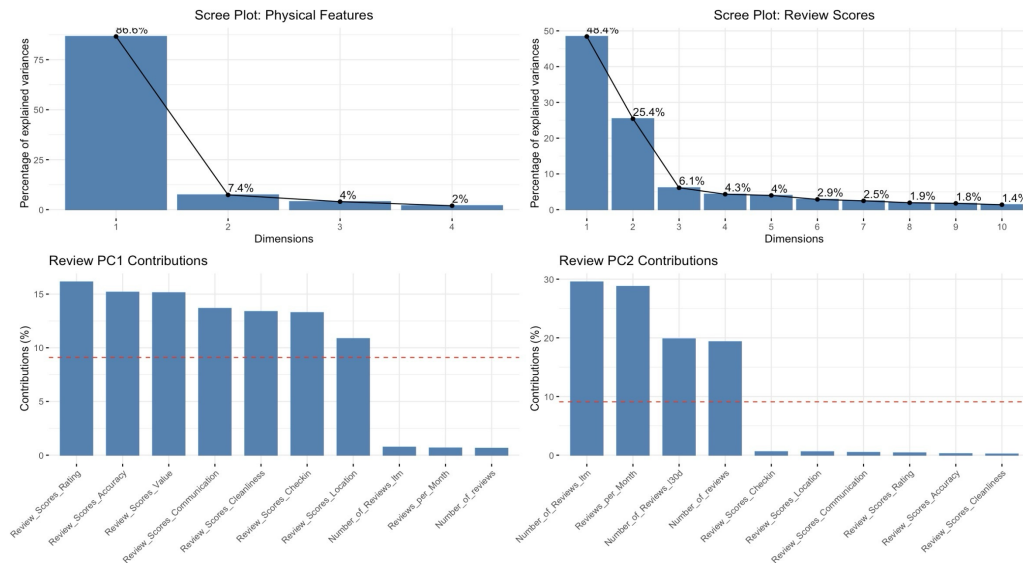


Figure 3: First row of plots shows the skree plots of two applications of PCA. Second row of plots shows the contributions of reviews variables on first two PCA components.

Table 2: Variables used further in the analysis

Variables	Description
Log_Price	The log of price variable
Room_Type	A factor containing 4 types of rooms.
Physical_Capacity	PCA component of related variables
Review_Quality	1st PCA component of review variables
Review_Volume	2nd PCA component of review variables

3 Cluster Analysis

Cluster Analysis was applied to identify the natural grouping among Cape Town Airbnb listings based on their physical and pricing characteristics.

3.1 Selecting Number of Clusters

Three criteria were used to select the number of clusters (K)

Elbow Method (WSS). The total within-cluster sum-of-squares was plottened against $K = 1, \dots, 10$. The elbow in Figure 4 suggests that gains from increasing the number of clusters becomes marginal after $K = 4$.

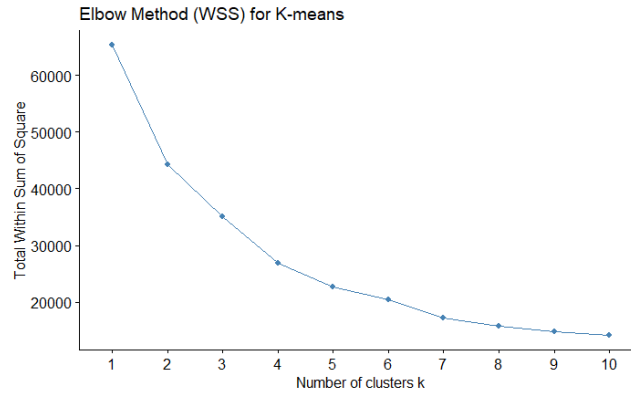


Figure 4: Elbow Method(WSS)

Average Silhouette Width. For each possible K, the mean silhouette width $\bar{s} = \frac{1}{n} \sum_{i=1}^n s_i$ was computed and plotted. Values near 1 indicate well separated clusters. The silhouette plot peaks at $K = 4$ as indicated in Figure 5.

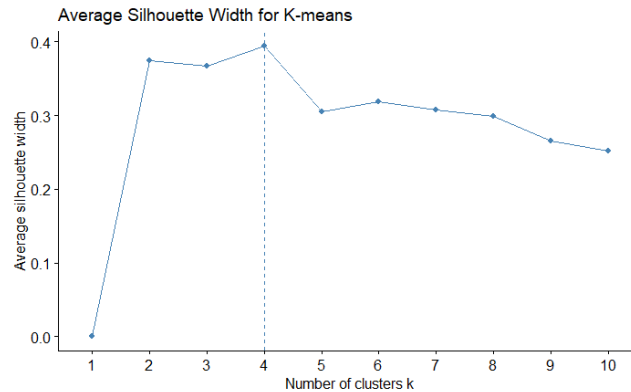


Figure 5: Ave Silhouette

Gap Statistic. The gap statistic compares the observed within-cluster variation to the expectation under a reference distribution. Figure 6 suggests that the smallest number of clusters after which the improvement becomes small relative to the uncertainty is $K = 3$.

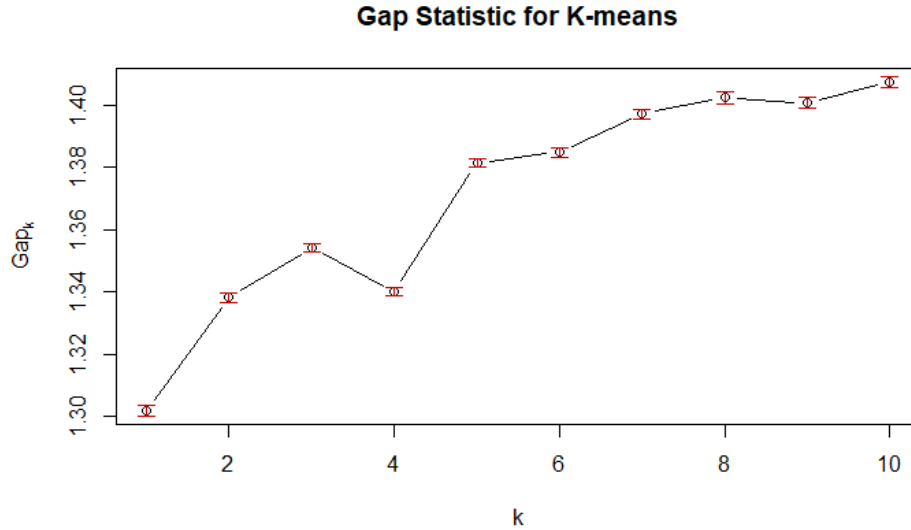


Figure 6: Gap Stat

Two of the methods suggested $K = 4$ was the most consistently supported choice to be used in all subsequent clustering analysis. The Gap statistic suggested $K = 3$ however we choose to go with $K = 4$ as it yields the most consistent results.

3.2 Hierarchical Clustering (HC)

Agglomerative hierarchical clustering was applied on training data. To account for the use of categorical variables the Gower dissimilarity coefficient was used. Complete linkage was used as it tends to produce compact clusters and is less prone to the chaining effect that characterises single linkage. The dendrogram in Figure 7 was cut at $K=4$, and the resulting clusters are outlined in the tree.

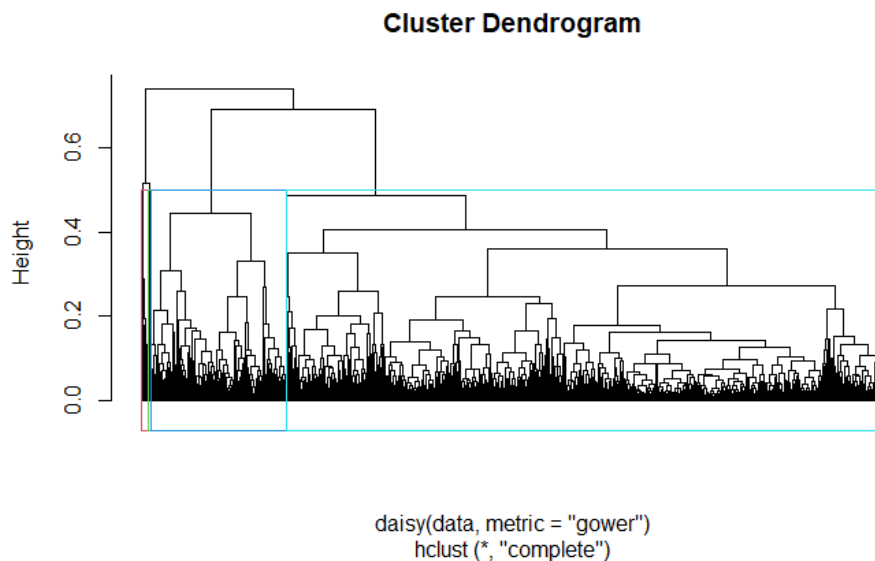


Figure 7: Dendrogram of hierarchical clustering solution. The tree is cut at $K=4$ clusters, indicated cluster outlines.

Cluster Profiles The coordinate plot in Figure 8 displays the standardised mean of each variable within each cluster in which the values are found in Table 6:

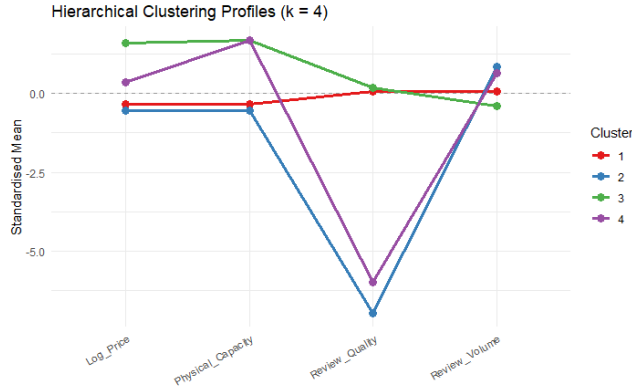


Figure 8: Hierarchical Clustering Cluster Profiles Plot

Table 3: K-means Cluster Summary (K = 4)

Cluster	n	Log Price	Phys. Capacity	Review Quality	Review Volume	Dominant Room Type
1	13,138	6.971	-0.552	0.128	0.154	Entire Home/Apt
2	150	6.770	-1.293	-13.883	1.093	Entire Home/Apt
3	2,972	8.519	2.454	0.405	-0.756	Entire Home/Apt
4	57	7.586	2.633	-14.130	1.094	Entire Home/Apt

Interpretation :

Cluster 1 - Standard Listings (n = 13,138): The dominant segment. It comprises of average physical capacity, review quality and, review volume with just below-average pricing. These are typical affordable properties with good service.

Cluster 2 -Poorly Reviewed Small Listings (n = 150): The cheapest cluster with a mean price of ($\exp(6.77) \approx R871$) and the smallest physical capacity. Unsurprisingly, due to the low price, this cluster has the second lowest review quality which signals consistently poor guest satisfaction.

Cluster 3 - Premium Large Listings (n = 2,972): The luxury segment. Highest prices ($\exp(8.52) \approx R5000$) and well-above-average physical capacity high-end properties. Review quality is above average, but review volume is below average, consistent with high-value properties that are booked less frequently.

Cluster 4 - Large Poorly Reviewed Listings (n = 57): Large properties at moderate-high prices yet low review quality with a high review volume. Indicating that unsatisfied guests.

Cluster Stability The stability of the hierarchical clustering solution was via bootstrap resampling. Figure 9. The mean ARI of 0.313 with a standard deviation of 0.132 indicates poor stability, with cluster assignments varying considerably across resamples. The distribution is broad and roughly centred around 0.3, spanning nearly the full range from 0.0 to 0.7, suggesting the hierarchical solution is sensitive to small perturbations in the data.

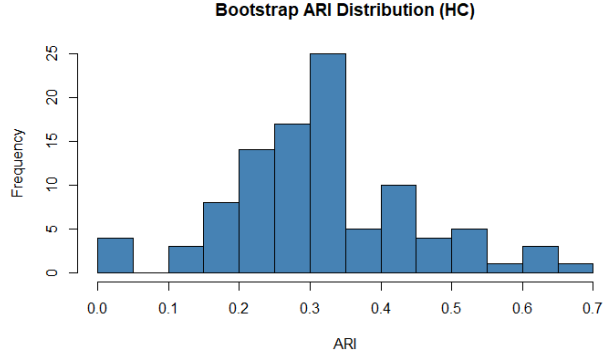


Figure 9: Bootstrap ARI distribution for the hierarchical clustering solution ($K = 4$)

3.3 Non-Hierarchical Clustering (NHC)

Two partition-based algorithms were applied: K-means and CLARA. Both used $K = 4$. K-means.

K-means partitions the n listings into K clusters by minimising the total within-cluster sum of squares. The algorithm alternates between assigning observation to its nearest centroid and recomputing the centroids until convergence.

CLARA is a medoid-based method that extends PAM to large datasets by applying PAM to a series of random subsamples. CLARA tends to be more robust to outliers than K-means

Cluster Visualisation Figures 10a and 10b project both solutions onto the first two principal components, which together explain 71.5% of total variance. The K-means solution shows four well-distributed clusters, with cluster 4 occupying the upper region and clusters 1–3 concentrated in the lower half. The CLARA solution follows a similar layout but cluster 2 is notably compressed and cluster boundaries overlap more.

Algorithm Selection The two solutions were compared using the Adjusted Rand Index (ARI) and mean silhouette width. An ARI of 0.462 reflects moderate agreement, indicating both algorithms recover similar broad structure while differing in finer cluster assignments. K-means achieved a higher mean silhouette width (0.393 vs. 0.272), suggesting more cohesive and better-separated clusters. This is also visible in Figures 10a and 10b, where the CLARA solution shows greater boundary overlap, consistent with its weaker silhouette score. Despite CLARA’s robustness advantage, K-means produced the stronger partition due to data processing that mitigated the affect of outliers and is adopted for all subsequent analysis.

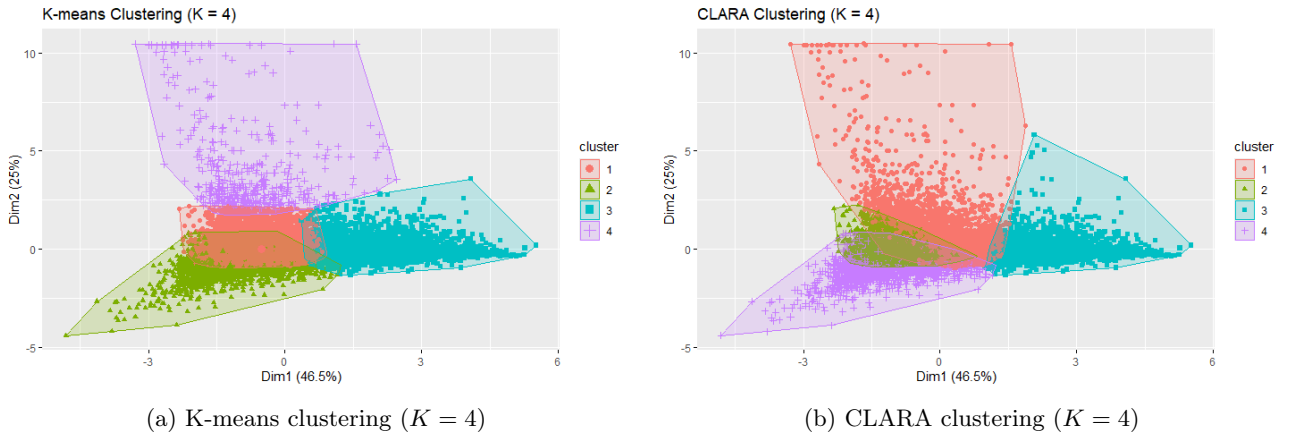


Figure 10: PCA projection of K-means and CLARA clustering solutions ($K = 4$). Dim1 and Dim2 account for 46.5% and 25% of total variance respectively.

Cluster Profiles The coordinate plot in Figure 11 displays the standardised mean of each variable within each cluster in which the values are found in Table 4:

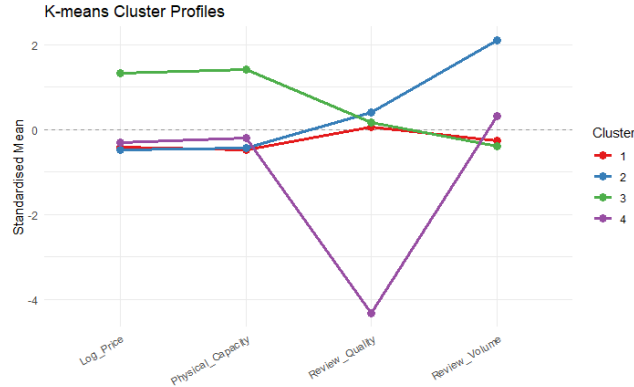


Figure 11: Mean standardised feature profiles for each K-means cluster ($K=4$).

Table 4: K-means cluster profiles ($K = 4$)

Cluster	n	Mean Log Price	Capacity	Review Quality	Review Volume	Room Type
1	9,988	6.893	-0.872	0.115	-0.439	Entire Home/Apt
2	1,926	6.852	-0.797	0.956	3.520	Entire Home/Apt
3	3,957	8.386	2.633	0.373	-0.665	Entire Home/Apt
4	446	6.990	-0.379	-10.012	0.528	Entire Home/Apt

Interpretation :

Cluster 1 -Standard Listings ($n = 9988$): The dominant segment, comprising listings with below-average physical capacity and review volume, average review quality, and modest pricing. These represent the typical affordable, adequately reviewed property.

Cluster 2 - High-Engagement Listings ($n = 1926$): High review volume and above-average review quality, suggesting these are frequently booked, well-regarded properties with Pricing and capacity that are comparable to Cluster 1, suggesting strong guest satisfaction drives repeat demand rather than price or size.

Cluster 3 - Premium Large Listings ($n = 3957$): The luxury segment. Highest mean log price ($\exp(8.39) \approx \text{R}4,400$) and well-above-average physical capacity.

Cluster 4 - The smallest cluster ($n = 446$), characterised by a severely negative review quality score. Suggesting guest dissatisfaction.

Cluster Stability The stability of the K-means solution was assessed via bootstrap resampling, with the Adjusted Rand Index (ARI) computed between the original and each resampled clustering. Figure 12 shows the resulting distribution. The mean ARI of 0.965 with a standard deviation of 0.018 indicates that the $K = 4$ solution is highly stable, with cluster assignments remaining largely consistent across resamples.

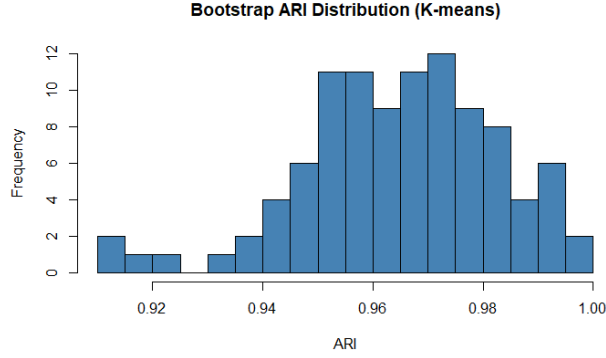


Figure 12: Bootstrap ARI distribution for the K-means clustering solution ($K = 4$)

3.4 Method Comparison

Table 5 summarises the performance of all clustering methods across mean silhouette width and bootstrap ARI stability.

K-means outperforms HC. It achieves the highest mean silhouette width (0.393), indicating more cohesive and better-separated clusters than HC (0.328). The ARI between HC and K-means of 0.31 reflects poor agreement, suggesting the two methods recover fundamentally different cluster structures. This is reinforced by the bootstrap stability results, where K-means yielded a mean ARI of 0.965 ($SD = 0.018$) compared to 0.313 ($SD = 0.132$) for HC, indicating that the K-means solution is highly replicable while the HC solution is sensitive to small perturbations in the data. Taken together, these results consistently favour K-means as the most reliable and well-defined clustering solution.

Table 5: Clustering method comparison across silhouette width and bootstrap stability.

Method	Mean Silhouette	Bootstrap ARI	Bootstrap SD
K-means	0.393	0.965	0.018
HC	0.328	0.313	0.132
CLARA	0.272	—	—

4 Self Organising Map

Self Organising Map (SOM) is a unsupervised neural network algorithm that aims to reduce high dimensionality data. It achieves this by constructing a map that preserves the structure of the data, which can then be visualised. After constructing the SOM map, a clustering algorithm can then be used to obtain clusters.

4.1 Training

A variation of SOM on the variables excluding ward average variables was conducted with supersom. Supersom allows different weightings of variables which was utilised to give different weights to categorical variables and numerical variables. Parameters used during training are as follows: 15x15 grid, rlen=2000, weights ratio of 1 numerical to 0.1 categorical and default alpha and radius. The resulting plots are shown in Figure 13.

The plots show that the parameters utilised are suited for current data. The training plot show that distances from each node's weights to the data samples represented by that node have flattened out. The counts plot is mostly uniform, which suggests that the grid size is not too large or too small. The U-matrix plot suggests natural boundaries at the edges of the matrix. The quality plot shows mostly small mean distance of objects mapped to respective nodes, suggesting a mostly well representation.

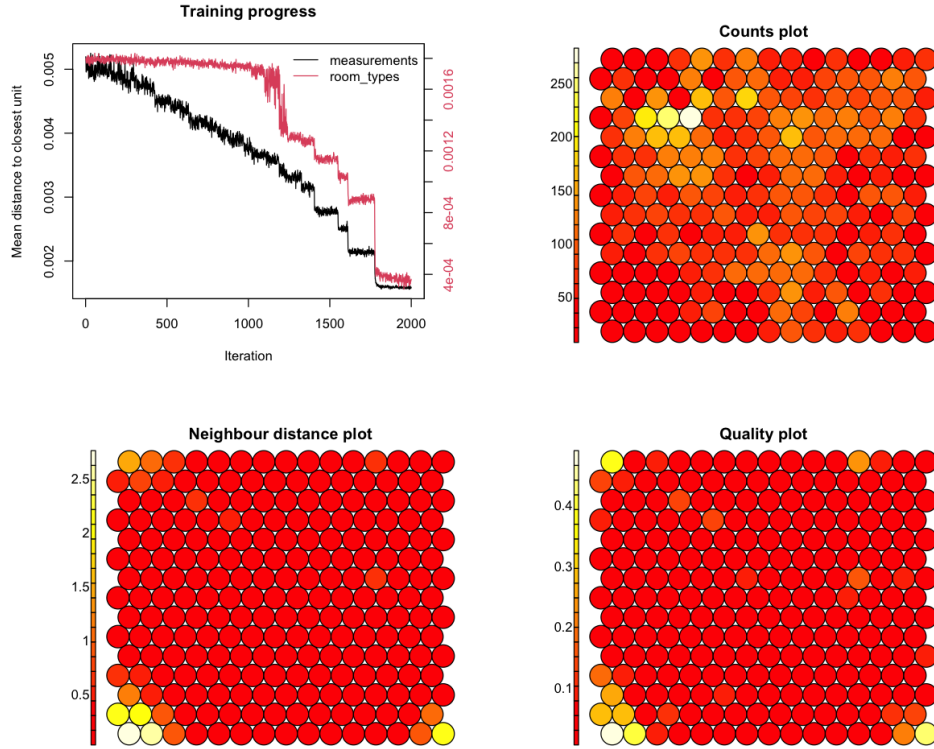


Figure 13: SOM Training and Map Plots

SOM heatmaps of variables are shown in Figure 14. The `log_price` and `physical_capacity` heatmap seem to correspond with high values near the bottom left and top right, with the low values in the middle having a rough hour glass shape. `Review_Quality` is mostly uniform, with low values at the bottom left. `Review_Volume` has a general gradient trend of low values from the left to high values at the bottom right. The mean silhouette of SOM clusters is 0.401, which indicates a modest clustering performance.

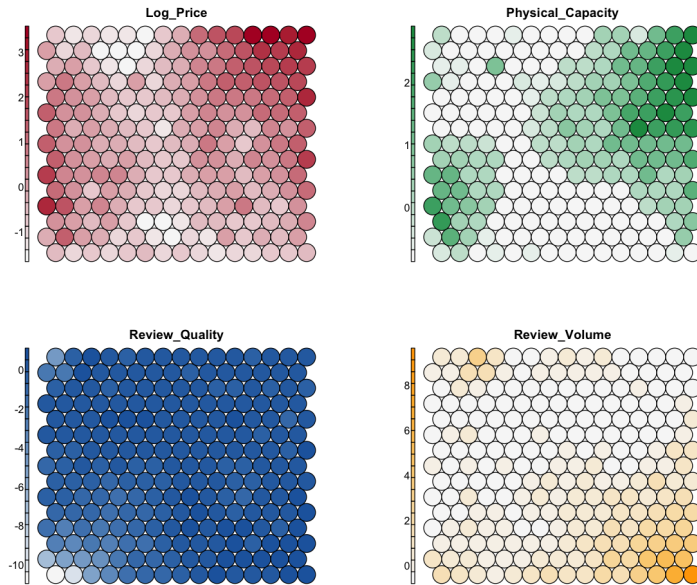


Figure 14: SOM Heatmaps of Numerical Variables

4.2 Cluster Analysis and Visualisation

K means clustering was performed on the weights of the SOM nodes to cluster the map. Plotting a WSS vs K shows a plot (Figure 15) that suggests 4 clusters with the elbow method.

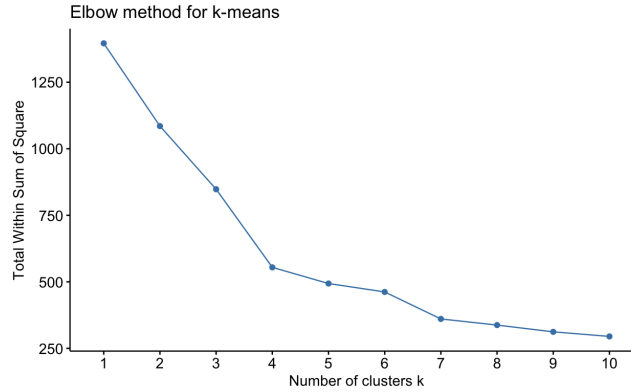


Figure 15: SOM K means elbow

The clusters are then mapped back to the SOM map visualisation which is shown in Figure 16. The clusters can also be plotted on a map in Figure 17. Previous clustering algorithms were also plotted on the map. However, the clusters seem to follow the same general trend as shown in Figure 17 of general interweaving of clusters. The density of properties is also more near the coastlines and Cape Town CBD.

4.2.1 Cluster 1: Poor Reviewed Properties

The properties in this cluster are categorised by their low reviews. They are commonly low in price, physical capacity. However, there is a node that indicates properties with high capacity and price with low reviews is also grouped here.

4.2.2 Cluster 2: Affordable Properties

The properties in this cluster are categorised by their affordable prices and average physical capacity. They have average review quality and volume. Most of the property listings are in this category. The properties in this cluster tend to occur all over Cape Town.

4.2.3 Cluster 3: Luxury Properties

The properties in this cluster are categorised by their high prices, physical capacity and high reviews. The high price makes results in not many people staying at the property and leaving a review, hence having a very small volume of review volume. The properties in this cluster tend to be present near the mountains and the coastline.

4.2.4 Cluster 4: Credible Properties

The properties in this cluster are categorised by their high volumes of high reviews. They are low in price and physical capacity. However, it seems that guests enjoy their stay at the Airbnb. Properties in this cluster tend to be present in dense areas with other property listings clusters. This cluster appears to be properties that are a good bargain for their price.

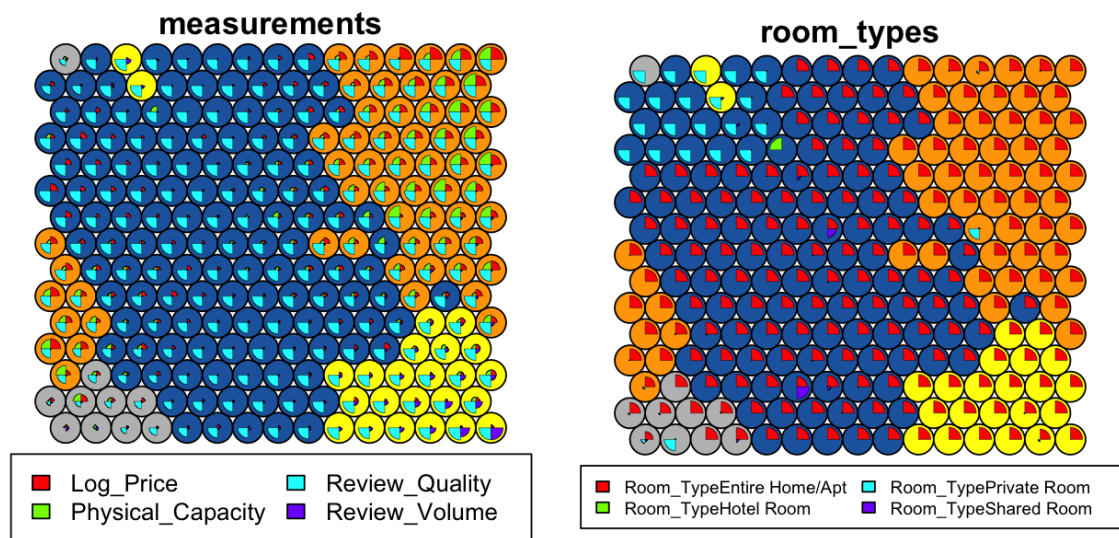


Figure 16: SOM Cluster Mapping: Cluster 1 (Grey), Cluster 2 (Blue), Cluster 3 (Orange), Cluster 4 (Yellow)

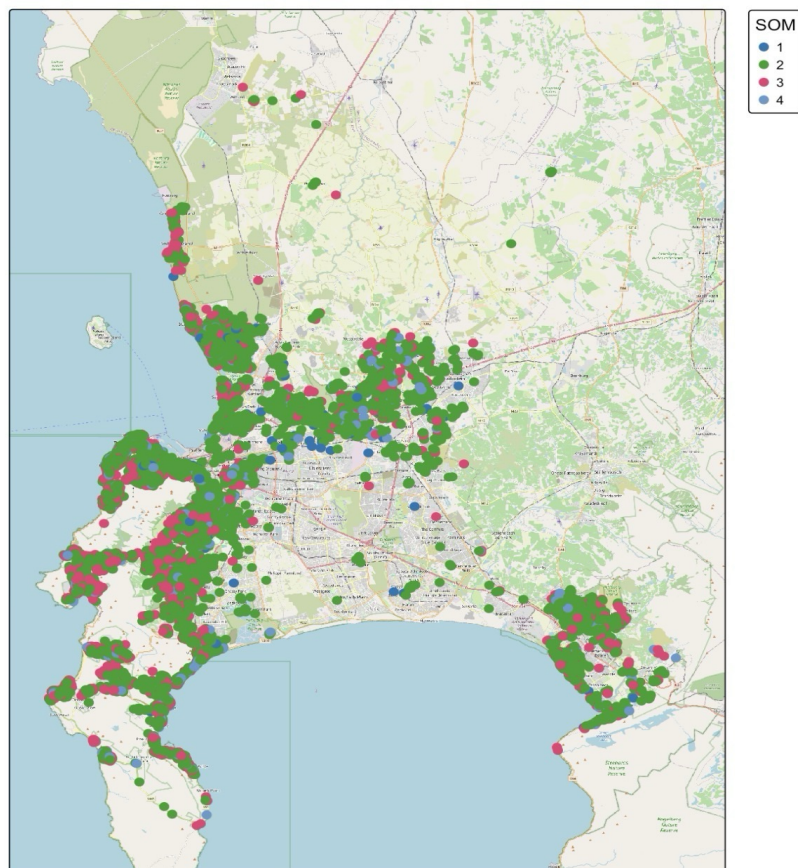


Figure 17: SOM Clusters on Map

4.3 Cluster Profiling

The information shows that the cluster profiles are similar to K-means algorithm. Values close to 0 (x-axis) represent values that are close to the average. The cluster profiles show the distinctive characteristics of each cluster as described in the previous section. (1) being poorly reviewed, (2) being average features, (3) being expensive and high physical capacity, (4) being high in review quality and volume.

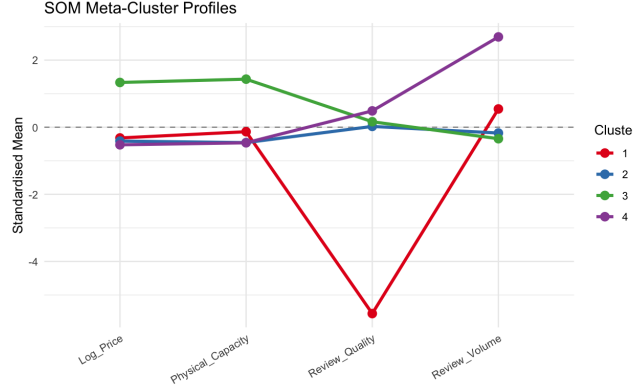


Figure 18: SOM Cluster Profiles

Table 6: SOM K-means Cluster Summary ($K = 4$)

Cluster	n	Log Price	Phys. Capacity	Review Quality	Review Volume	Dominant Room Type
1	265	6.981	-0.249	-12.817	0.911	Entire Home/Apt
2	10,990	6.903	-0.853	0.056	-0.290	Entire Home/Apt
3	3,911	8.387	2.667	0.379	-0.573	Entire Home/Apt
4	1,151	6.808	-0.863	1.124	4.502	Entire Home/Apt

4.4 Comparison

The Self-Organising Map (SOM) with K-mean clustering of the SOM Map yielded insightful results. The visualisation of cluster on the map makes gives a more informative visualisation than standard cluster profiling.

The cluster profiles of K-means and SOM are very similar. Which suggests that SOM and K-mean clusters are capturing the same structure in the dataset. Additionally, the silhouette score of SOM (0.401) improves on K-mean's silhouette score (0.393). This suggests that SOM provides a structure which allows the K-means to better capture the underlying clusters.

Both the base K-means and the SOM K-means models converged on identical overarching market types:

- **Affordable/Standard Properties:** The biggest market segment, characterised by average physical capacities, modest pricing, and average reviews.
- **Credible/High-Engagement Properties:** Listings that achieve above-average review quality and high review volumes despite maintaining low prices and smaller physical capacities.
- **Luxury/Premium Properties:** High-priced, large-capacity properties which have lower review volumes due to fewer bookings.
- **Poorly Reviewed Properties:** A minority of properties identified by negative review scores and guest dissatisfaction.

5 Conclusion

This report aimed to explore the underlying structures of Airbnb properties in Cape Town, with a specific focus on understanding the guest experience across different geographical zones. Through comprehensive exploratory data analysis and dimensionality reduction via Principal Component Analysis (PCA), the dataset was condensed into key variables representing physical capacity, review quality, review volume, and pricing.

Multiple unsupervised learning techniques, including Hierarchical Clustering, CLARA, K-means, and Self-Organising Maps, were applied and evaluated. The K-means algorithm proved to be highly stable compared to HC and CLARA. However, the combination of SOM followed by K-means clustering proved to be the optimal algorithm, achieving the highest mean silhouette width of 0.401.

The analysis reveals that the Cape Town Airbnb market is heavily dominated by affordable, standard properties dispersed widely across the city. However, some geographical structure exists. Luxury, high-capacity properties are commonly clustered near the coastline and mountainous regions, while highly credible, frequently reviewed properties tend to concentrate in denser urban areas. It is found the guest experience is largely positive across geographical locations, except for a small minority of poorly reviewed properties.